Indefinite Integration

Mind map

Basics

- $\int f = \phi + C \iff \phi' = f$
- Given value fixes C

Standard results

- Five sets of results
- $ax + b$: divide by a
- Simplify first

1/Quad, $1/\sqrt{\text{Quad}}$, $\sqrt{\text{Quad}}$

- Complete the square
- $\tan^{-1}$, ln, $\sin^{-1}$ forms

Substitution

- $\phi(x) = t$ when ϕ' present
- $\int f'/f = \ln|f|$
- $\ln x = t$, $\tan^{-1} x = t$

Trig substitution

- $a^2 + x^2 \to a\tan\theta$
- $a^2 - x^2 \to a\sin\theta$
- $x^2 - a^2 \to a\sec\theta$

Integration by parts

- ILATE
- 1 as second function
- Cyclic: $e^{ax}\sin bx$

Classic integrals

- $e^x(f + f') \to e^x f$
- $f + xf' \to xf$
- $e^g(fg' + f') \to e^g f$

Rational functions

- Divide if improper
- linear $= A\,Q' + B$

Partial fractions

- Distinct: $\frac{A}{x-a}$
- Repeated: up to $\frac{A_k}{(x-a)^k}$
- Quadratic: $\frac{Bx+C}{Q}$

$\sin^m x \cos^n x$

- Odd: other $= t$
- Both even: multiple angles
- $m + n$ neg. even: $\tan x = t$

Other trig forms

- $\sin^2, \cos^2$ in Dr: $\tan x = t$
- $\sin, \cos$ in Dr: $\tan\frac{x}{2} = t$
- Nr $= A\,\text{Dr} + B\,\text{Dr}' + C$
- $\sin x \pm \cos x$ with $\sin 2x$

Miscellaneous

- Kutur putur: $1 + x^{-n} = t$
- $\frac{x^2 \pm k}{x^4 + px^2 + k^2}$: $x \mp \frac{k}{x} = t$
- $\sqrt{x^2} = |x|$

Irrational functions

- $(x-\alpha)(\beta - x)$: $\alpha\cos^2\theta + \beta\sin^2\theta$
- $\sqrt{px+q} = t$; $ax + b = \frac{1}{t}$; $x = \frac{1}{t}$
- $m + n = 2$: $\frac{x-a}{x-b} = t$